Nahush Shrivatsa

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Education

New York University — September 2025 – May 2027
Master of Science in Computer Science — *Cumulative GPA: 3.7*

University of Washington, Seattle Campus — September 2021 – June 2025
Bachelor of Science in Computer Science — *Cumulative GPA: 3.8 (Annual Dean's List)*

Technical Skills and Certifications

Languages: Java, Python, OCaml, JavaScript, C, C++, TypeScript, SQL, HTML/CSS, YAML, Bash
Technologies: React Native, JSON, Kubernetes, Docker, Flask, MongoDB, Figma, Git, Red Hat OpenShift

Relevant Experience

NYU IT High Speed Research Network — New York, NY
Research Engineer — *January 2026 – Present*

- Research Engineer on the real-time applications team, collaborating with labs across NYU to design and deploy high-throughput computational pipelines on HSRN infrastructure.
- Building a **CalmAn + Fiola Python** pipeline for the Fenton Lab that processes neuronal fluorescence imaging data to neural activity signals in real-time, building a temporal dataset & enabling closed-loop dynamic experimentation.

Nokia — Dubai, UAE
Software Engineer Intern II — *June 2024 – September 2024*

- Developed Customer Experience Index solution integrating KPI data with **MongoDB** and a **Python** Flask app, generating Grafana dashboards and increasing analysis productivity by 60%.
- Led system administration of **OpenShift** platform; managed users, maintained technical documentation, and deployed 10+ containerized workloads and 4 **GitLab** CI/CD pipelines.
- Engineered a Gen AI application reducing document analysis time by 70% using **LLaMA3**, **RAG**, and a vector database to scan documents and auto-generate domain-specific testing scenarios.

Software Engineer Intern I — *July 2023 – October 2023*

- Implemented a scalable distributed file store with **GPG encryption** using **Apache NiFi** and **Bash scripting**, resulting in a 60% increase in memory performance and throughput.
- Built an autonomous file sync system between private cloud and SharePoint via **Python** scripts and **API calls**, yielding a **100% improvement** in document management operations.

Systems Neuroscience & AI Lab (Golub Lab) — Seattle, WA
Undergraduate Researcher — *September 2024 – June 2025*

- Collaborated with Dr. Matt Golub and Dr. Jacob Sacks on modeling population-level neural activity from motor tasks with **recurrent neural networks** to probe circuit-level control strategies.
- Built a modular SLURM-friendly **PyTorch + Hydra** platform fusing custom RNNs with biomechanical simulations, providing researchers a scalable plug-and-play backend on HPCs.

Additional Experience

Washington Tunneling — Seattle, WA
Lead Software Engineer | Innovating tunnel boring machines | GitHub — *January 2024 – June 2025*

- Spearheaded **C++** firmware integration on an ESP32 microcontroller with multiple sensors and modules, enabling real-time data transmission and implementing critical control functionalities.
- Developed robust serial communication protocol to process and package sensor data and commands between the microcontroller and **Python** server, managing machine dynamics and GUI data feeds.

Paul G. Allen School of Computer Science and Engineering — Seattle, WA
Undergraduate Teaching Assistant — *March 2024 – June 2025*

- TA for Programming Languages (3+ quarters: types, scope, semantics, ASTs in **OCaml, Ruby, Racket**) and Systems Architecture (pointers, caches, x86 assembly) to 75+ students. Delivered guest lectures to 80+ students.

Projects

Home Server | *Docker, WireGuard, Caddy, Bash* | GitHub — October 2025

- Architected and self-host a **multi-service home server** on a Virtual Machine running 15+ **Docker** containers across isolated network stacks, including Jellyfin, Seafile, and Vaultwarden.
- Authored Daemon & Bash automation scripts for 0 downtime rolling updates, image cleanup & instant Tailscale link.

Distributed Systems Lab | *Java, Paxos, Sharding, RPC* | GitHub — May 2023

- Developed a sharded, linearizable, highly available key-value store with dynamic load balancing using Paxos in Java.